

A Hybrid LSTM (Deep Learning)-LLM Framework for 10-Year Treasury Yield Forecasting

Yeonjun Lee

Korea International School, Gyeonggi, Republic of Korea

I. ABSTRACT

Traditional models for forecasting the 10-Year U.S. Treasury Yield are limited by their reliance on quantitative data, often failing to capture market sentiment from financial news [1][2]. This research enhances prediction accuracy by engineering a sentiment feature using a large language model (LLM) and a role-based prompt framework. In a controlled experiment, a Long Short-Term Memory (LSTM), a type of deep learning model, augmented with this sentiment score was compared to a baseline model using only traditional economic indicators. The integrated model demonstrated significantly superior performance, reducing Root Mean Squared Error (RMSE) by **17.4%** and Mean Absolute Error (MAE) by **24.6%**. These results validate that quantifying news sentiment with an LLM provides a powerful predictive signal for financial time-series forecasting.

II. METHODS

This study's methodology was designed to evaluate the efficacy of a hybrid Deep Learning-Large Language Model (DL-LLM) architecture. The process involves training the model on a historical dataset of T-bond yields and then comparing the model's predictions with the actual data to determine the architecture's significance.

A. Testing Method

A controlled experiment was designed to measure the impact of the sentiment feature. Two identical Long Short-Term Memory (LSTM) models [3] were constructed and trained using the TensorFlow framework:

Model A: A **baseline model** using only the quantitative data.

Model B: An **experimental model** augmented with the LLM-generated sentiment feature.

The predictive performance of both models was then rigorously evaluated on a held-out test set using **Root Mean Squared Error (RMSE)** and **Mean Absolute Error (MAE)** metrics. The **Diebold-Mariano test** was used to determine if the performance difference was statistically significant [4].

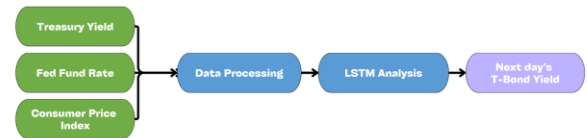
B. Dataset and Timeframe

The dataset spans from **January 1, 2009, to December 31, 2020**, a period selected for its comprehensive news data availability and economic significance. This 12-year window provides a consistent post-2008 crisis training regime that captures a full economic cycle and the COVID-19 pandemic, offering a robust range of market conditions from stable growth to extreme volatility. Quantitative data, such as the 10-Year Treasury Yield, Federal Funds Rate, and CPI (Consumer Price Index), were sourced from Yahoo Finance and FRED, while qualitative data was provided by a Kaggle dataset of daily financial news headlines[5][6][7].

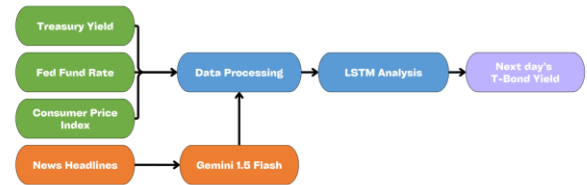
III. EXPERIMENTAL DESIGN

Figure 1. A Visual Overview of the Experimental Methodology for the Baseline and Integrated Models.

Model A: Baseline Architecture (LSTM Analysis Only)



Model B: Integrated Architecture (LSTM Analysis + LLM-Generated Sentiment)



To test the hypothesis that an LLM-generated sentiment score provides predictive value, we designed a controlled experiment using two parallel models, as illustrated in Fig. 1.

- **Model A (Baseline Architecture LSTM Analysis Only):** This model represents the traditional, quantitative approach to financial forecasting. It was trained using only three historical time-series features: the 10-Year Treasury Yield, the Federal Funds Rate, and the CPI. By excluding any qualitative data, this model serves as a benchmark, establishing a performance baseline that reflects the capabilities of a standard LSTM network on purely econometric data.
- **Model B (Integrated Architecture LSTM Analysis + LLM-Generated Sentiment):** This model represents the proposed hybrid approach. It was trained on the same three quantitative features as Model A but was augmented with the fourth, LLM-generated sentiment score. The purpose of this model is to test whether the inclusion of this sentiment feature can explain variance

¹Yeonjun Lee is with Korea International School, Gyeonggi, Republic of Korea (e-mail: Yjuni0810@gmail.com, Ylee27@studnet.kis.or.kr, Yeonjun@uchicago.edu)

in the Treasury Yield that is not captured by the historical quantitative data alone.

By isolating the sentiment feature as the sole independent variable between the two models, any statistically significant difference in their predictive performance can be confidently attributed to the value of the sentiment score.

Justification for Choosing a Controlled Experiment: Our controlled experiment was deliberately chosen to isolate the sentiment score's impact. Other approaches were unsuitable: a single model with all features would not measure the sentiment's specific contribution, and comparing different architectures (like LSTM vs. ARIMA) would make it impossible to know if performance gains came from the feature or the model design. Our parallel design is the only method that cleanly and directly tests the study's central hypothesis.

IV. MODEL ARCHITECTURE AND FEATURE ENGINEERING

The core innovation was the creation of the sentiment score [8]. Daily news headlines were submitted to the Gemini 1.5 Flash LLM. Its analytical behavior was guided by a detailed prompt instructing it to act as a "meticulous financial analyst" and adhere to a strict rubric:

- **Focus:** Concentrate only on macroeconomic news related to U.S. inflation, employment, and Federal Reserve policy.
- **Ignore:** Disregard company-specific earnings, stock market chatter, and irrelevant political news.
- **Scoring:** Provide a single decimal score from -1.0 (Extremely Dovish, indicating a weakening economy or falling inflation) to +1.0 (Extremely Hawkish, indicating high inflation and a robust economy).

The Long Short-Term Memory (LSTM) network serves as the predictive engine of this study, specifically engineered to forecast the next day's T-bond Yield by analyzing complex time-dependent patterns within the prepared dataset. The entire process, from data preparation to model compilation, was meticulously configured for this time-series forecasting task.

Data Preparation for Time-Series Forecasting

Before training, the data was prepared using a two-step process. First, all input features were normalized to a uniform range of 0 to 1 to ensure stable and efficient learning. Second, this data was transformed into 30-day lookback windows, creating samples where 30 consecutive days of data are used to predict the Treasury Yield of the 31st day.

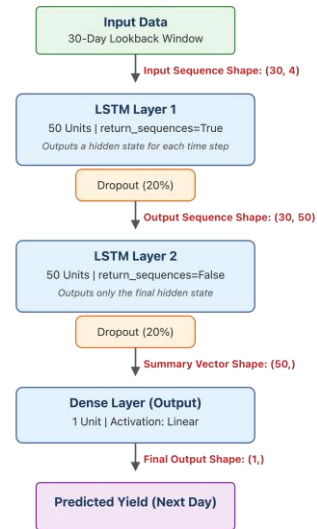
LSTM T-Bond Yield Forecasting Model Architecture

The model's architecture, detailed in Fig. 2 is constructed as a Sequential model with two stacked LSTM layers, a design

proven effective for financial market prediction [9], to learn hierarchical temporal representations from the data.

Figure 2. A Visual Overview of the Experimental Methodology for the Baseline and Integrated Models.

LSTM T-Bond Yield Forecasting Model Architecture Flow



Input Sequence and Data Flow

The input for the LSTM model is a 3D tensor with a shape of **(30, 4)**. This structure is specifically designed for time-series forecasting and represents:

- **30 Time Steps:** A **30-day lookback window** was chosen as it aligns with a standard one-month period of trading data. This timeframe is long enough to capture meaningful short-term trends and market patterns without being so long that it includes outdated, irrelevant information.
- **4 Features:** At each time step, the model receives four key indicators: the 10-Year Treasury Yield, the Federal Funds Rate, the Consumer Price Index (CPI), and the LLM-generated sentiment score.

As this input sequence passes through the network, its shape is transformed:

1. The initial **(30, 4)** input sequence enters the first LSTM layer.
2. This layer outputs a sequence of 50-dimensional hidden states, resulting in an **Output Sequence** with a shape of **(30, 50)**.
3. The second LSTM layer processes this new sequence and outputs only the final hidden state, creating a **Summary Vector** with a shape of **(50,)**. This vector is a dense representation of the entire 30-day period.

The model's architecture consists of a sequential network specifically designed to handle time-series data. Key components include stacked LSTM layers to learn temporal patterns, dropout layers to prevent overfitting, and a dense output layer to produce the final prediction.

- **Stacked LSTM Layers:** The architecture uses two LSTM layers, each with **50 units**. The "units" refer to the dimensionality of the hidden state, which is the memory vector that captures information from previous time steps. Stacking the layers allows the first layer to learn basic temporal patterns, while the second layer can learn more abstract patterns from the sequence of outputs from the first.
 - The **first LSTM layer** is configured with `return_sequences=True`. This is a critical setting that instructs the layer to output the hidden state for *every time step* in the 30-day sequence, not just the final one. This provides the full sequence of learned features to the next layer.
 - The **second LSTM layer** uses the default `return_sequences=False`. It processes the entire sequence of hidden states from the first layer and outputs only a single vector representing the hidden state of the very last time step. This single vector acts as a dense summary, effectively encoding the most important information from the entire 30-day period.
- **Regularization and Output:** To prevent overfitting, a **Dropout layer with a 20% rate** is applied after each LSTM layer. During training, this randomly sets 20% of the input units to zero at each update step, forcing the network to learn more robust and generalizable features. Finally, this summary vector is passed to a **Dense layer**, which acts as a regression head to map the learned features to a single, continuous output value: the predicted Treasury Yield.

Training and Compilation

To guide the learning process, the model was compiled using the 'adam' optimizer for its known efficiency and the 'mean_squared_error' (MSE) loss function to penalize large prediction errors. The model was then trained for 50 epochs to ensure convergence on a predictive solution.

V. RESULTS AND DISCUSSION

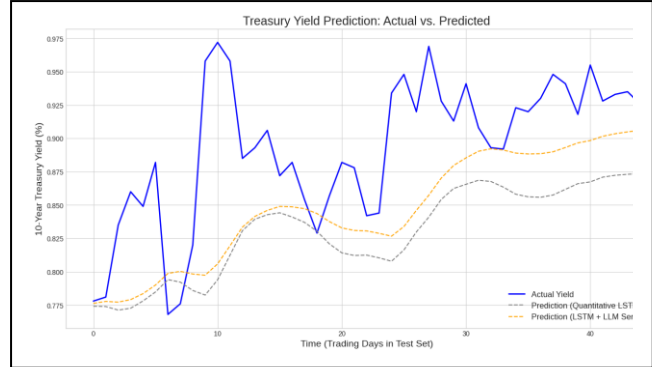
Evaluation on the held-out test set reveals a significant performance improvement from the sentiment-integrated model (Model B). As detailed in Table 1, Model B outperformed the baseline (Model A) by reducing the Root Mean Squared Error (RMSE) by 17.4% and the Mean Absolute Error (MAE) by 24.6%. A Diebold-Mariano test confirms this result is statistically significant ($P < 0.0001$), allowing us to reject the null hypothesis of equal predictive accuracy between the two models.

TABLE I. COMPARISON OF MODEL PERFORMANCE ON THE TEST SET.

TABLE I	Model Performance Comparison		
	Features	RMSE ^a	MAE ^b
Model A	Quantitative Only (LSTM)	0.0765	0.0647
Model B	Quantitative + Sentiment (LSTM + LLM)	0.0632	0.0488
Reduction in Error		17.4%	24.6%

a. RMSE: Root Mean Squared Error

Figure 3 A comparison of the actual 10-Year Treasury Yield against the predictions from the baseline model (Quantitative LSTM) and the integrated model (LSTM + LLM Sentiment) over the held-out test set.



The quantitative improvement is confirmed in Fig. 3. The success of the integrated model stems from its ability to process a forward-looking indicator; while CPI and Federal Funds Rate data are lagging, news sentiment captures the market's real-time expectations. This gives Model B superior responsiveness, allowing it to track volatile market movements more dynamically, whereas the baseline model's predictions appear dampened and slow to react.

Despite its success, the model has clear limitations. At times, its predictions overshoot market peaks, suggesting an overreaction to headlines that may have already been "priced in" by the market. This highlights a subtle difference between a headline's stated sentiment and its ultimate market impact. Furthermore, the study's reliance on news headlines rather than the full text of articles is a primary limitation that offers an avenue for future work.

This research successfully demonstrates that integrating an LLM-engineered sentiment feature significantly enhances the predictive accuracy of a neural network for forecasting the 10-Year U.S. Treasury Yield. The work validates a powerful hybrid framework for blending unstructured, qualitative data with traditional quantitative analysis, opening a new frontier for more robust and responsive financial forecasting tools.

REFERENCES

- [1] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed. Hoboken, NJ: Wiley, 2015.
- [2] P. C. Tetlock, "Giving content to investor sentiment: The role of media in the stock market," *The Journal of Finance*, vol. 62, no. 3, pp. 1139–1168, 2007.
- [3] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

- [4] F. X. Diebold and R. S. Mariano, "Comparing predictive accuracy," *Journal of Business & Economic Statistics*, vol. 13, no. 3, pp. 253–263, 1995.
- [5] Yahoo Finance, "^^TNX," [Online]. Available: <https://finance.yahoo.com>. [Accessed: June. 14, 2025].
- [6] U.S. Federal Reserve, "Federal Reserve Economic Data (FRED)," Federal Reserve Bank of St. Louis, [Online]. Available: <https://fred.stlouisfed.org>. [Accessed: June. 14, 2025].
- [7] A. Misleading, "Massive Stock News Analysis & NLP Backtests," Kaggle, 2020, [Online]. Available: <https://www.kaggle.com/datasets/miguelaelle/massive-stock-news-analysis-db-for-nlpbacktests>. [Accessed: June. 14, 2025].
- [8] D. Araci, "FinBERT: Financial Sentiment Analysis with Pre-trained Language Models," presented at the 19th Int. Conf. on AI in Finance, New York, NY, USA, 2019.
- [9] T. Fischer and C. Krauss, "Deep learning with long short-term memory for financial market predictions," *European Journal of Operational Research*, vol. 270, no. 2, pp. 654-669, 2018.